

DS 201: Data Handling and Visualization

Course Code: DS 201

Course Name: Data Handling and Visualization

L-T-P-C: 2-0-2-3

Prerequisites: None

Intended for: UG

Distribution: Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines

Semester: Even/Odd

Approval: 29th BoA (B.Tech DSE curriculum approved by the 20th Senate); re-approved in the 55th BoA for the DSE 2022 and earlier batches; credits recorded as 3 (corrected from 4) in the 65th BoA; removed from the Second Major CSE list as closed in the 65th BoA

Preamble

This course will provide undergraduate students with an understanding of the data collection, data representation and data visualization. Furthermore, this course will provide a general framework that allows students to identify errors associated with different data sources. Students will learn some metrics to quantify each potential error, and thus they will be able to describe the quality of a data source. The course will introduce different large-scale data collection efforts undertaken by industry and government agencies, and review the learned concepts through case-studies and examples. This course is suitable for students who are looking for a general framework to use data sources.

Objective

With the help of various examples students will learn how to identify which data sources likely matches research questions, how to turn research questions into measurable pieces, and how to think about an analysis plan.

On completion of this course, the student should be able to

- describe what makes a good or bad visualization
- demonstrate familiarity with various data sources
- Identify the functions that are best for particular problems
- Understand best practices for creating basic charts
- Learn data analysis tools and software

Course Modules

- **Data sources and collection:** This module walks you through the process of data collection. Starting with a review of existing structured and unstructured data sources, we cover data collection techniques using sensors, surveys, and different instruments. This includes data collection and storing for different domains such as IoT, Audio and Video, Web and Social Networks etc. (6 lectures)
- **Data Pre-processing:** Highlight the importance of data correction and discuss some basic features that can affect your data analysis when dealing with sample data. Issues of data access and resources for

access are introduced in this module. Descriptive data summarization, data cleaning, normalization, data integration and transformation, data reduction. (7 lectures)

- **Data representation:** Importance of data representations, Extracting salient features from data, Examples include MFCC from audio signals, histogram representation for text, feature representations for images, encoded representations, Spatial data representation: cartography, GIS paper maps to ArcGIS ArcMap symbolizing, Time-series data representations and curve fitting. (9 lectures)
- **Basic charting and data visualization:** Basic charting, examples with real world weather data, extract and manipulate the data to display the maximum information, various types of graphs like pie chart, bar graphs, 3-D plots using Matlab and R. Procedure of composite charts by overlaying a scatter plot of record breaking data for a given year, Visualization of high dimensional data e.g. TSNE plot, histogram etc. Also, dynamic data representations and visualization of data using D3.js (6 lectures)

Lab Exercises

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should be appropriately be sequenced for synchronization with the laboratory.

Text Books

1. Yau, Nathan. *Visualize this: the Flowing Data guide to design, visualization, and statistics*. John Wiley & Sons, 2011.
2. Tufte, Edward R. *The visual display of quantitative information*. Vol. 2. Cheshire, CT: Graphics press, 2001.

Reference Books

1. Janert, Philipp K. *Data analysis with open source tools: a hands-on guide for programmers and data scientists*. O'Reilly Media, Inc., 2010.
2. Zhu, Xuan. *GIS for environmental applications: a practical approach*. Routledge, 2016.